Deep Learning Enabled Design of RF/mmWave IC and Antennas

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***Abstract*—Design of RF and mmWave circuits and systems is a tedious process that requires an expert team to balance out many requirements and trade-offs. This requires numerous sub-tasks to be assigned to different engineers, who then design based on their knowledge, intuition and experience. On the other hand, we are observing a paradigm shift in many other fields, thanks to artificial intelligence (AI). It is not only possible to automate many tasks, but AI can come up with solutions on demand. In this regard, there is ample reason to investigate AI assisted methods for design and automation of RF and mmWave circuits and systems. This paper introduces an algorithmic synthesis approach that relies on deep convolutional neural networks (CNN) for the modelling of template-free electromagnetic (EM) structures. It is worth noting that once a CNN model is trained, synthesis time is measured within minutes. Moreover, this model can be repeatedly used for different design targets, such as antennas, matching networks, and filters. These points are exemplified with synthesis and measurement results.**

***Index Terms*—mmWave, 5G, inverse design, machine learning, SiGe, power amplifier, antenna**

I. Introduction

Electromagnetic structures, along with transistors, are the backbone of RF/mmWave circuits. Especially at higher frequencies, maximum gain that can be extracted from devices are limited, and impedance mismatches between subsequent stages degrade performance. Historically, design of EM structures has relied on intuition, trial-and-error, and parameter sweeps. As such, synthesizing, tuning and verifying EM structures could be time consuming and iterative. In parallel to these, automating the EM synthesis has seen a surge of interest [1]–[3]. However, a fixed topology with a constrained parameter space may not be optimal in terms of efficiency or area. In this regard, a new design space allowing arbitrary shapes could potentially yield better performing results [4]–[9]. As shown in Fig. 1-a, for a given set of target S-Parameters, inverse design tries to synthesize an EM structure without relying on an initial template or manual human intervention. While it is desirable to have arbitrary shapes, due to manufacturability concerns, a pixelated grid was chosen to comply with design rules (Fig. 1-b). These pixelated grid structures can offer exponentially large

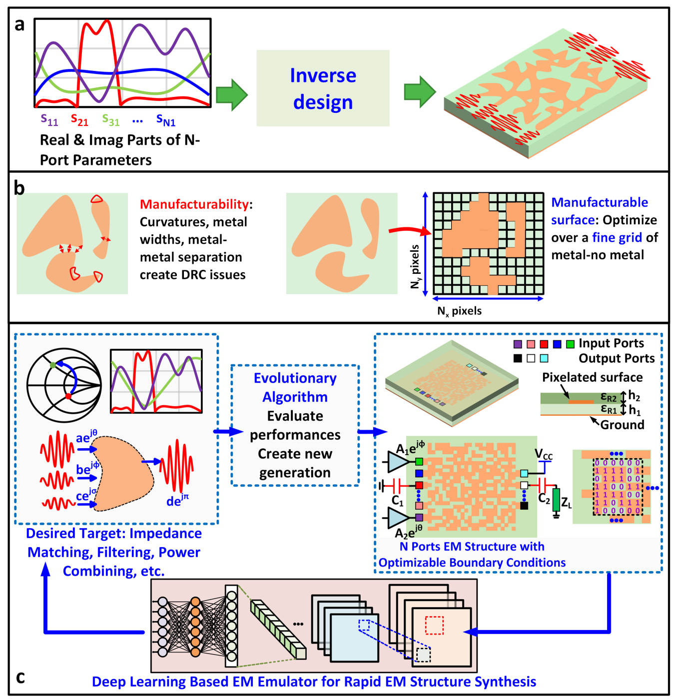

Fig. 1. (a) Given the targeted multi-port S-Parameters, inverse design algorithmically attempts to find a solution without relying on a template. Consequently, results can often be non-intuitive and unattainable reachable with classical synthesis approaches. (b) While fully arbitrary surfaces are the ideal outcome of inverse design, manufacturability and design rule constraints requires an approximation. A convenient approximation to arbitrary surfaces would be creating a grid like structure where each pixel is a design freedom. (c) Due to large number of design freedoms, optimization via repeated EM simulations is prohibitively resource intensive. Instead, we train a convolutional neural network to predict the S-Parameters of a pixelated structure. This CNN is then used with an optimization algorithm to synthesize desired EM characteristics. Given the discrete nature of the design space, population algorithms, such as genetic algorithm, are well suited for optimization.

design freedoms as each pixel can be chosen to be metal or no metal. However, given the non-convexity of the problem, navigating this large design space by relying on iterative

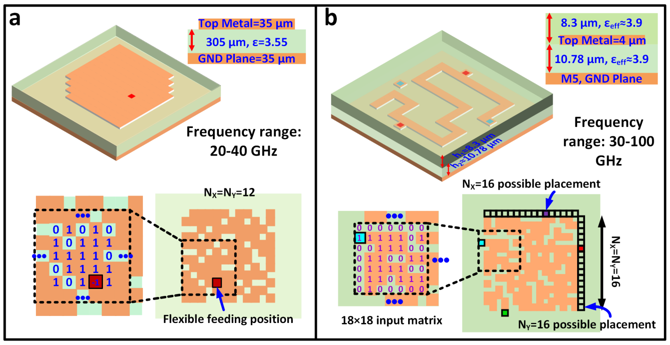

Fig. 2. (a) Design space and comparison for a single-port antenna structure embedded in a high-frequency dielectric between 20-40 GHz, with a flexible feeding position (12×12 pixels). (b) Design space for multi-port, arbitrary structures between 30-100 GHz (16×16 pixels). In either cases, discrete matrix representation of these EM structures are input to the CNN based EM predictor.

EM simulations is extremely challenging. To address this, a deep learning model (CNN) that can predict S-Parameters and radiation characteristics was trained as shown in Fig. 1-c. Consequently, optimization could be ran on the space of near-arbitrary structures without performing iterative EM simulations, which are replaced with CNN predictions. More specifically, modelling and synthesis of 2-port on-chip EM structures [6] and off-chip PCB antennas [7] are described in this work, as shown in Fig. 2. Once the training is done, CNN based optimization could be performed repeatedly for different design goals. Section–II will detail the deep learning based modelling and inverse synthesis algorithm with design examples. Section–III will also present measurement results in support of these findings.

II. Deep Learning Enabled Inverse Design of RF/mmWave EM Structures

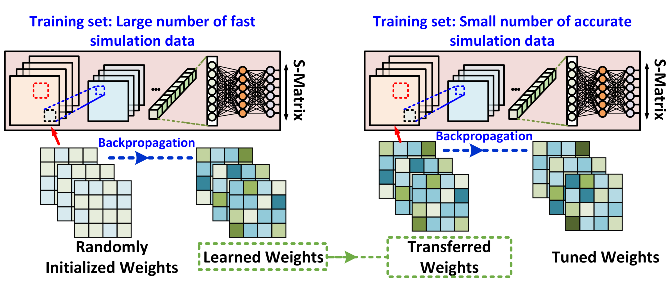

Fig. 3. Transfer learning is applied to allow efficient training of the EM predictor. The initial training is done through stochastic gradient descent on a large sample of coarsely meshed simulation data. Learned weights are then used as the starting point for a second set of training on a smaller sample of more accurate simulations. This provides better accuracy and training time compared to training done only with the latter accurate simulation data.

A. Accurate and Rapid Modelling of Near Arbitrary EM Structures via Deep Learning

The EM emulator for rapid prediction is the enabling component of the inverse design approach, as shown in Fig. 1-c. Given the image-like representation of pixelated structures, convolutional networks are the natural choice for extracting the EM characteristics. Despite being powerful tools for modelling, training of CNNs requires sizable datasets. Moreover, high pixel count required for near-arbitrary structures necessitates a relatively large number of training samples, which are obtained through EM simulations. In order to reduce simulation resources, two step transfer learning was applied [10]. In the first step, large number of randomly generated structures were simulated with coarse mesh (hence reduced simulation time). A randomly initialized CNN was then trained with this simulation data. Even though data used in this step does not have the desired granularity and accuracy, the model can still extract relevant features. More importantly, a subsequent smaller set of simulations with better accuracy (finer mesh) then proves sufficient to tune this initial model. This is conceptually illustrated at Fig. 3. Another important aspect for training of accurate models is optimizing the layer combination. As shown in Fig. 1-c, the EM predictor model consists of convolutional and fully connected (FC) layers [11]. Each convolutional and FC layer was followed by batch normalization and a leaky ReLU activation function. In addition, FC layers have dropout operation to counteract overfitting. Number of convolutional and FC layers were optimized through parameter sweeps [6].

B. Design Examples

Once training is complete, the model can be deployed in conjunction with an optimization algorithm. Due to the ease of implementation and ability to tackle local minimas, population algorithms are well suited to the non-convex optimization we are performing. As it is shown in Fig. 2, population members in a generation are denoted via discrete valued matrices, which is also the preferred input format for CNNs. With the S-Parameters or radiation characteristics predicted, each member was evaluated with a predefined cost function. Creation of the next generation is then performed according to the cost values, i.e., better performing members have a better chance for being selected as parents [6]. As an example, a population size of 4096 was evolved over 100 generations. Thanks to the rapid prediction of EM simulation results, this optimization only takes minutes. Moreover, the cost function can be changed on demand for different design goals. Fig. 4 exemplifies the generalizability aspect of inverse synthesis. In all 4 cases, the same CNN model was utilized while the optimization cost function was changed. We can see that, vastly different targets were successfully synthesized. Fig. 4-a shows 70 GHz band pass and notch filters, whereas Fig. 4-b shows broadband PA matching network for different device sizes.

Furthermore, it is possible to apply the same principles of the inverse design method in different domains, albeit with newly generated datasets and different CNN models. The second case we consider in this work is RF/mmWave antenna synthesis. As it was shown in Fig. 2-a, antennas were synthesized for a standard PCB substrate. Different from the on-chip case, feeding position could be placed anywhere

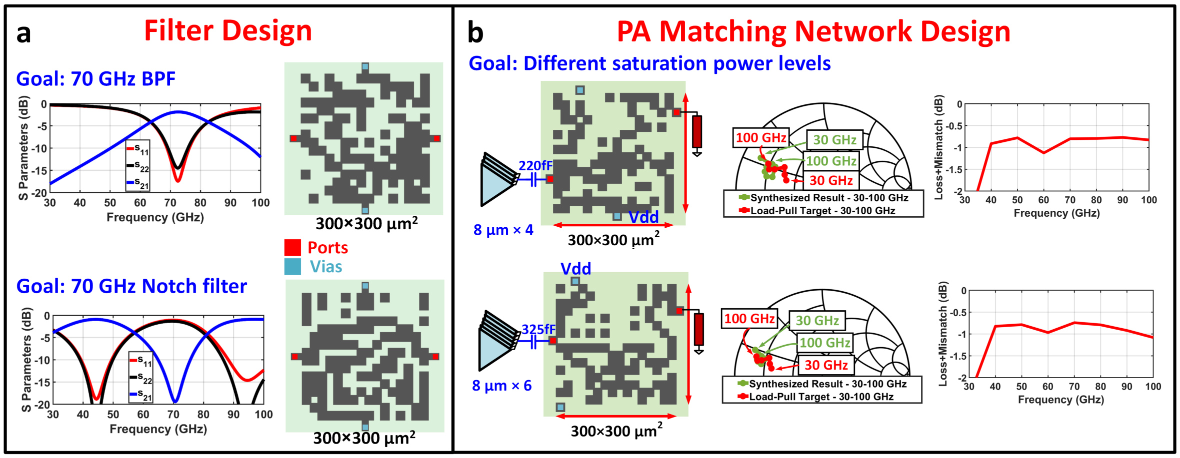

Fig. 4. Sample designs generated through the proposed inverse-design methodology using one trained model. (a) A band-pass filter and a notch filter are synthesized targeting 70 GHz, with extracted S-parameters shown. The structure dimension is 300×300 μm. (b) Matching networks are synthesized for different PA cell sizes, targeting a wideband match from 30-100 GHz. The extracted structures are compared to the ideal load-pull targets at each frequency, showing minimal loss/mismatch over the band. In addition to matching, these networks are able to integrate the DC feed ports.

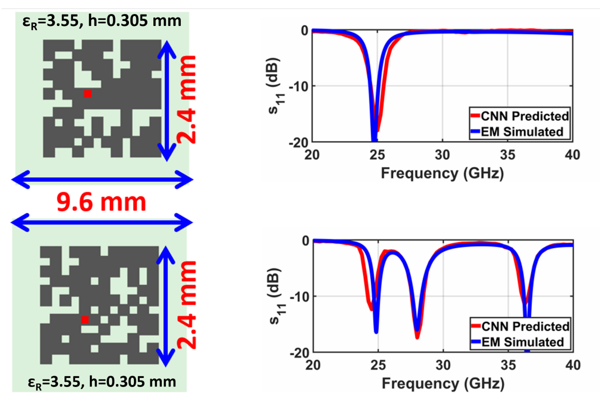

Fig. 5. Inverse-designed compact antennas. A single-frequency antenna centered at 25 GHz is synthesized, with another example targeting triple-band operation at 25, 28, and 37 GHz.

within the antenna surface. Fig. 5 exemplifies synthesis results with compact and multi-band patch antennas. It is noteworthy that a square patch with 2.4 mm edges would resonate at 30 GHz, whereas inverse design was able to synthesize solutions that can operate below this frequency. In addition, multi-band antennas are easily synthesized, compared to the considerable effort required with classical design methods. In all the example designs, synthesis time is confined to less than 5 minutes. With a GPU, EM prediction is almost instantaneous. In fact, most of the synthesis time is spent on different overheads, such as cost function evaluation and off-spring generation [6].

III. Measurement Results

In parallel to the generalizability aspect of the inverse design method, we applied the synthesis loop for a common base (CB) power amplifier cell output matching network (Fig. 6). Details of this work can be found in [6]. When compared with common emitter, CB presents elevated mmWave gain and

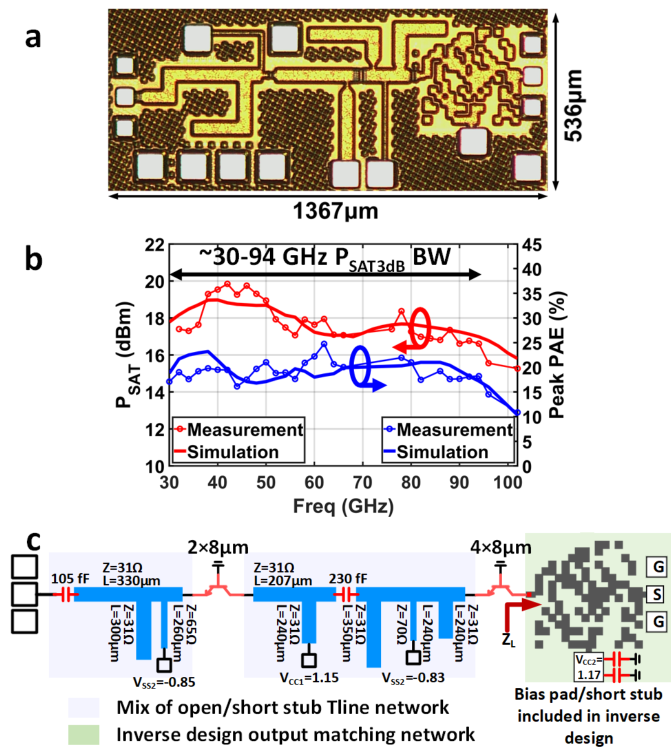

Fig. 6. (a) A 30-94 GHz SiGe PA supporting concurrent multiband operation was demonstrated with an inverse-design output matching network. [6] (b) P_{sat} and PAE versus frequency is shown, with measurement and simulation results compared. PA achieves 16.7-19.5 dBm P_{sat} across 30-94 GHz. (c) Both driver and PA cells are in CB base configuration. To avoid stability problems, base is grounded. Input and interstage matching networks consist of multi-order open/short stub transmission line networks.

breakdown voltage. As a result of these, 30-100 GHz range can be potentially covered simultaneously. However, to enable this coverage, matching network designs need to be carried out with great care. Thanks to the low loss, broadband, inversely designed output matching network, the PA achieves 30-94 GHz P_{Sat3dB} BW (16.7-19.5 dBm) with PAE of 16-24.7% in the same range. Moreover, concurrent dual band and triple band modulation with aggregate data rate reaching 7.5 Gbps were demonstrated.

Lastly, measurement results for antenna synthesis are shown [7]. Here we show a compact RF antenna (Fig. 7-a) and a wide-band mmWave antenna (Fig. 7-b). The RF antenna was designed for 60 mil RO4003C substrate. With the given bounding box size, a typical patch antenna would resonate around 4 GHz while the inverse design solution is resonant at 2.5 GHz, showing enhanced area efficiency. When compared to a traditional patch antenna of same size, Fig. 7-b, enhances the matching BW by utilizing 2 resonances around 30 GHz. While S-Parameter measurements match well, due to non-idealities in the radiation pattern measurement (such as bulky edge connectors for mmWave antennas and reflections from the environment for RF antennas), we observe mismatch between simulation and measurement. Nevertheless, beam characteristics qualitatively correlate in both cases.

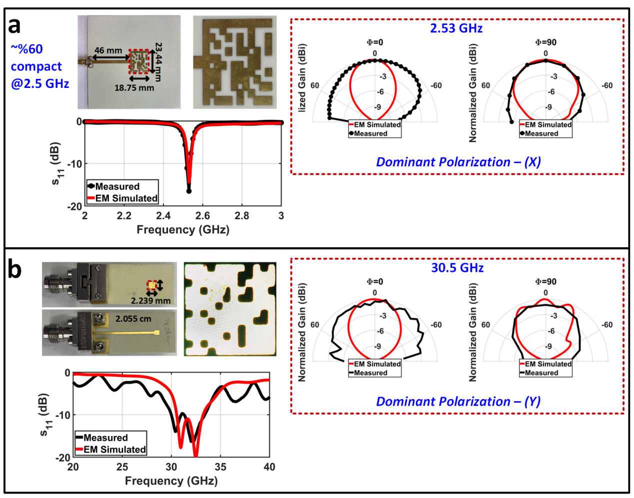

Fig. 7. (a) A compact, single-frequency antenna at 2.5 GHz with edge feed is shown. (b) A broadband antenna is shown, with measurement and simulation data compared.

IV. Conclusion

In this article, we present an inverse-design methodology for generating arbitrary EM structures based on desired S-Parameter or radiative responses. This allows for rapid search and generation of more optimal designs outside the confines of traditional template-based structures. A deep-learning based forward model is trained as an EM simulation emulator, accurately predicting scattering parameters of arbitrary structures. Example designs demonstrate filters, matching networks, and antennas for various applications. The proposed technique allows for rapid, automated synthesis of complex circuits and systems with new functionalities.

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